Intro. to 2-D vectors (平面向量)

In mathematics and physics, quantities can be classified into two major types:

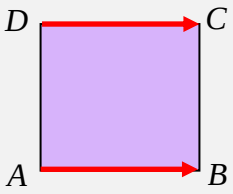
1. Scalar (純量) : quantities with magnitude only
   e.g. mass *m*, area *A*, distance *s*, temperature *T*, etc.
2. Vector (向量) : quantities which have both magnitude and direction
   e.g. velocity $\vec{v}$, weight $\vec{w}$, force $\vec{f}$, etc.

In this notes, it will cover the basic concepts of 2-D vectors (i.e. vectors in $\mathbf{R}^2$).

Basic definition of vectors

(a) Two vectors are equal if they have the same magnitude and direction.

e.g. In the figure, $\overrightarrow{AB} = \overrightarrow{DC}$.



A square with side 2 unit

(b) It two vectors have same magnitude but opposite in direction, they will differ by negative sign.

e.g. In the figure, $\overrightarrow{AB} = -\overrightarrow{BA}$.

(c) Zero vector (null vector), denoted by $\vec{0}$, is a vector with zero magnitude

e.g. In the figure, $\overrightarrow{AA} = \vec{0}$.

(d) The magnitude of any vector $\vec{u}$ is denoted by $|\vec{u}|$.

e.g. In the figure, $\left|\overrightarrow{AB}\right| = 2$.

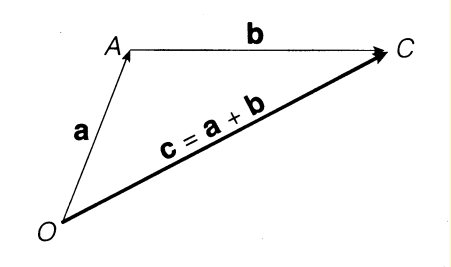
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Addition and Subtraction of vectors

Addition of vectors

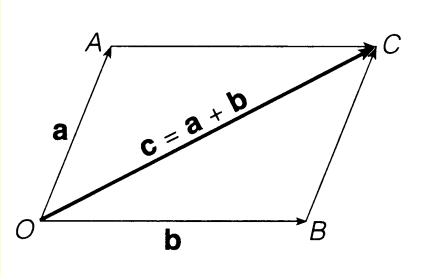
Addition of two vectors can be illustrated by Triangle law or Parallelogram law:

1. Triangle law



$$\overrightarrow{OA}+\overrightarrow{AC} = \overrightarrow{OC}$$

2. Parallelogram law



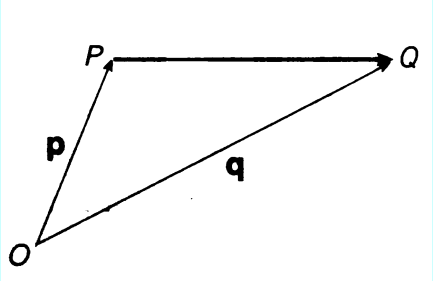
$$\overrightarrow{OA}+\overrightarrow{OB} = \overrightarrow{OC}$$

(Note: In general $\overrightarrow{OA}+\overrightarrow{AC} = \overrightarrow{OC}$ but $\left|\overrightarrow{OA}\right|+\left|\overrightarrow{AC}\right| \neq \left|\overrightarrow{OC}\right|$.)

Subtraction of vectors

Subtraction of two vectors can be illustrated by Triangle law:

- **Triangle law**



$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ}$$

i.e. $\overrightarrow{OQ} - \overrightarrow{OP} = \overrightarrow{PQ}$

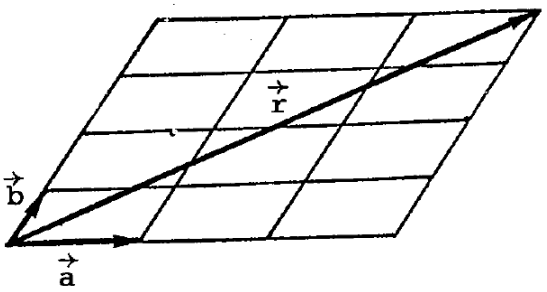
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Components of vectors

If there are two **non-zero and non-parallel vectors** $\vec{a}$ and $\vec{b}$, then we can consider them as **base vectors** and resolve other vectors as linear combination of $\vec{a}$ and $\vec{b}$ in the vector space (向量空間).

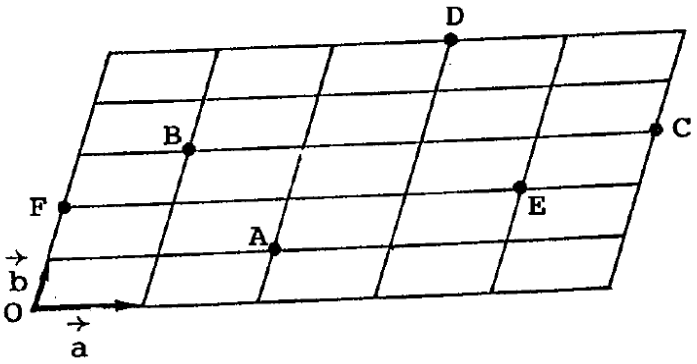
e.g. In the figure, we have

$\vec{r} = 3\vec{a} + 4\vec{b}$



Let's TRY IT!

Refer to the figure below, express the following vectors as linear combination of $\vec{a}$ and $\vec{b}$.



(a) $\overrightarrow{OA}$ (b) $\overrightarrow{OC}$ (c) $\overrightarrow{OE}$

(d) $\overrightarrow{OF}$ (e) $\overrightarrow{BA}$ (f) $\overrightarrow{DE}$

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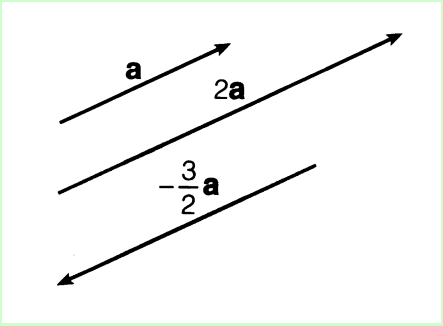
Scalar multiplication of vectors

Scalar multiplication of vectors

When a vector $\vec{a}$ is multiplied by a scalar m, the product $m\vec{a}$ is a vector parallel to $\vec{a}$ such that

(a) The magnitude of $m\vec{a}$ is m times that of $\vec{a}$.

(b) When $m > 0$, $m\vec{a}$ has the same direction as $\vec{a}$,
When $m < 0$, $m\vec{a}$ has the opposite direction of $\vec{a}$.



As a summary, here is some basic properties of operation of vectors:

Basic Properties of vectors

1 Properties of vector addition

For any two vectors **a** and **b**, we have

(a) $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$

(b) $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$

2 Properties of scalar multiplication

Let m, n be two non-zero scalars. For any two vectors **a** and **b**, we have

(a) $m(n\mathbf{a}) = (mn)\mathbf{a} = n(m\mathbf{a})$

(b) $(m + n)\mathbf{a} = m\mathbf{a} + n\mathbf{a}$

(c) $m(\mathbf{a} + \mathbf{b}) = m\mathbf{a} + m\mathbf{b}$

(Note : In the textbooks / notes, vectors are usually represented by $\vec{a}$, and sometimes by **a** (bold letters only) for convenience.)

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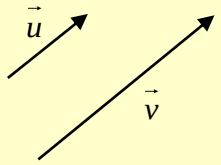
Parallel vectors

Parallel vectors

For any **non-zero vectors** $\vec{u}$ and $\vec{v}$,

$\vec{u}$ and $\vec{v}$ are **parallel** (or **collinear**) if and only if

$\vec{v} = k\vec{u}$ where k is non-zero constant

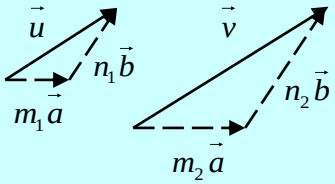


Parallel vectors from addition of non-parallel vectors

For any **non-zero and non-parallel vectors** $\vec{a}$ and $\vec{b}$,

Suppose there are non-zero scalars m_1, m_2, n_1 and n_2 :

$\vec{u} = m_1\vec{a} + n_1\vec{b}$ and $\vec{v} = m_2\vec{a} + n_2\vec{b}$



then

(a) $\vec{u}$ and $\vec{v}$ are **parallel** (or **collinear**) if and only if $\dfrac{m_1}{m_2} = \dfrac{n_1}{n_2}$

(b) $\vec{u}$ and $\vec{v}$ are **equal** if and only if $m_1 = m_2$ and $n_1 = n_2$

e.g. For two **non-zero and non-parallel vectors** $\vec{a}$ and $\vec{b}$,

- $2\vec{a} + 3\vec{b}$ and $4\vec{a} + 6\vec{b}$ are parallel;
- $2\vec{a} + 3\vec{b}$ and $3\vec{a} + 2\vec{b}$ are not parallel

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Parallel vectors

[Add. Math 2003 Q.6]

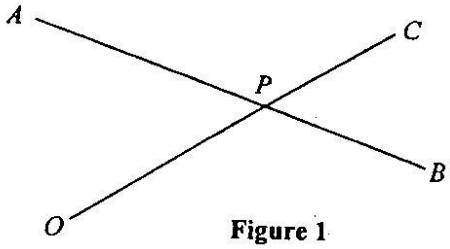


Figure 1

In Figure 1, point P divides both line segments *AB* and *OC* in the same ratio 3 : 1.

Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$.

(a) Express $\overrightarrow{OP}$ in terms of **a** and **b**.

(b) Express $\overrightarrow{OC}$ in terms of **a** and **b**.
Hence show that *OA* is parallel to *BC*. ***(5 marks)***

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Unit vectors

Unit vector

Unit vector is a vector with **magnitude equals to 1**

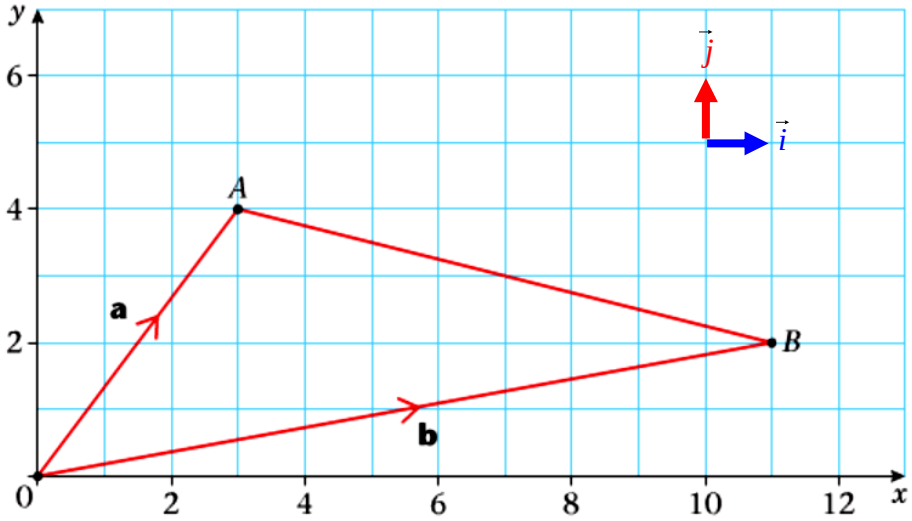
For any vector $\vec{a}$, its unit vector, denoted as a, can be found as follows:

$$\text{unit vector } a = \frac{\vec{a}}{|\vec{a}|}$$

Unit vectors in 2-D rectangular coordinate plane

In the *x*-*y* coordinate plane, there are two **important unit vectors**, $\vec{i}$ and $\vec{j}$:

- $\vec{i}$ represents the unit vector in the **+ve direction of *x*-axis**
- $\vec{j}$ represents the unit vector in the **+ve direction of *y*-axis**



e.g. In the figure, we can expressed $\overrightarrow{OA}$ and $\overrightarrow{OB}$ into components $\vec{i}$ and $\vec{j}$:

- $\overrightarrow{OA} = 3\vec{i} + 4\vec{j}$ and $\left|\overrightarrow{OA}\right| = \sqrt{3^2 + 4^2} = 5$
- $\overrightarrow{OB} = 11\vec{i} + 2\vec{j}$ and $\left|\overrightarrow{OB}\right| = \sqrt{11^2 + 2^2} = \sqrt{125}$

(Remark: $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are called **position vectors** of *A* and *B* respectively.)

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Vectors in 2-D rectangular coordinate plane

Let $P\ (x\ ,\ y)$ be any point in the 2-D coordinate plane. The **position vector** $\overrightarrow{OP}$ defines the position of P relative to the origin O of the coordinate plane, where

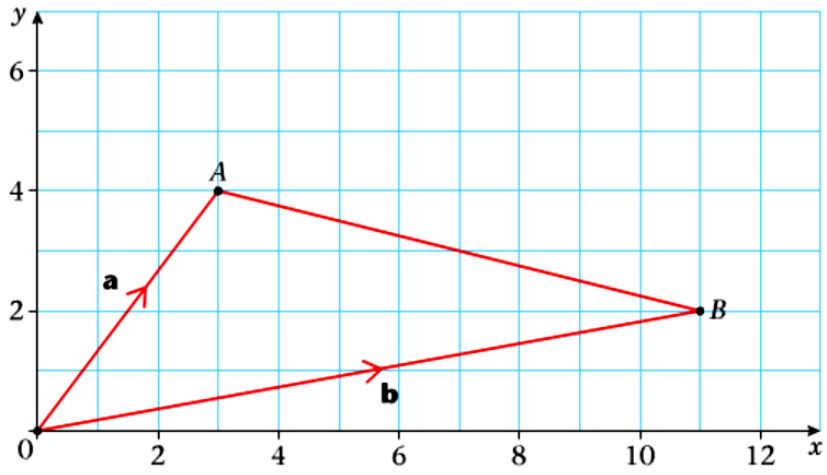
$$\overrightarrow{OP} = x\vec{i} + y\vec{j} \qquad \text{and} \qquad \left|\overrightarrow{OP}\right| = \sqrt{x^2 + y^2}\ .$$

Furthermore,

If $A = (x_1\ ,\ y_1)$ and $B = (x_2\ ,\ y_2)$ be any two points in the 2-D coordinate plane,

then $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (x_2 - x_1)\ \vec{i} + (y_2 - y_1)\ \vec{j}$

and $\left|\overrightarrow{AB}\right| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}\ .$



e.g. In the figure, $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$$= (11-3)\vec{i} + (2-4)\vec{j}$$

$$= 8\vec{i} - 2\vec{j}$$

and $\left|\overrightarrow{AB}\right| = \sqrt{8^2 + (-2)^2} = \sqrt{68}\ .$

Therefore the unit vector in the direction of $\overrightarrow{AB}$ is

$$\frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|} = \frac{8}{\sqrt{68}}\vec{i} - \frac{2}{\sqrt{68}}\vec{j}$$

Note : Position vector is also known as **line-localized vectors**, which are different to **free vectors** which can be moved freely.

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Vectors in 2-D rectangular coordinate plane

Concept checking:

Let $\mathbf{u} = 5\mathbf{i} + 12\mathbf{j}$ and θ be the acute angle that $\mathbf{u}$ made with the x-axis. Determine whether each of the following is true.

1. $|\mathbf{u}| = 13$ T / F
2. $\tan\theta = \frac{5}{12}$ T / F
3. θ lies in quadrant I. T / F
4. The unit vector of $\mathbf{u}$ is $\frac{5}{13}\mathbf{i} + \frac{12}{13}\mathbf{j}$. T / F
5. If $\mathbf{v}$ is a vector having the same direction as $\mathbf{u}$ and $|\mathbf{v}| = 2$,
   then $\mathbf{v} = \frac{10}{13}\mathbf{i} + \frac{24}{13}\mathbf{j}$. T / F

The points A, B and C have coordinates (3, –1), (4, 5) and (–2, 6) respectively, and O is the origin.

(a) Find, in terms of **i** and **j**,

(i) the position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$,

(ii) $\overrightarrow{AB}$.

(b) Are A, B and C collinear? Explain your answer.

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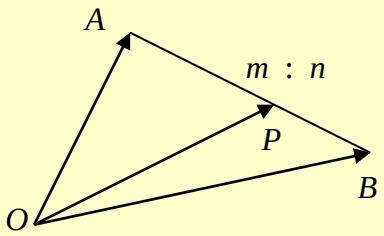
Section Formula

Section formula in vectors

For any points A and B with another point O,

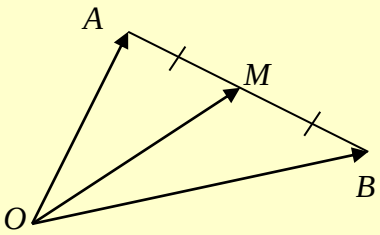
(a) if $AP : PB = m : n$,

$$\overrightarrow{OP} = \frac{n\overrightarrow{OA} + m\overrightarrow{OB}}{m+n}$$



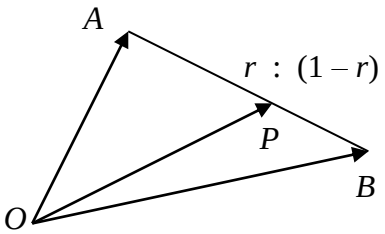
(b) if M is the **mid-point** of AB,

$$\overrightarrow{OM} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2}$$



Note: If we let $AP : PB = r : (1 - r)$, then we have

$$\mathbf{p} = (1 - r)\mathbf{a} + r\mathbf{b}$$



where **p**, **a** and **b** are position vectors of P, A and B respectively. Furthermore,

(i) P coincides A for $r = 0$,
(ii) P coincides B for $r = 1$,
(iii) P lies between A and B for $0 < r < 1$,
(iv) P lies on BA produced for $r < 0$,
(v) P lies on AB produced for $r > 1$.

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Section Formula

[Add. Math 2000 Paper I Q.8]

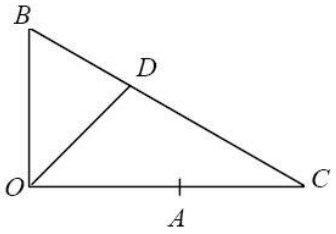


Figure 1

In Figure 1, $OA = \mathbf{i}$, $OB = \mathbf{j}$. C is a point on OA produced such that $AC = k$, where $k > 0$. D is a point on BC such that $BD : DC = 1 : 2$.

(a) Show that $\overrightarrow{OD} = \frac{1+k}{3}\mathbf{i} + \frac{2}{3}\mathbf{j}$.

(b) If $\overrightarrow{OD}$ is a unit vector, find k. ***(4 marks)***

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HKCEE Questions (level 3)

1. ***[Add. Math 2001 Q.14]***

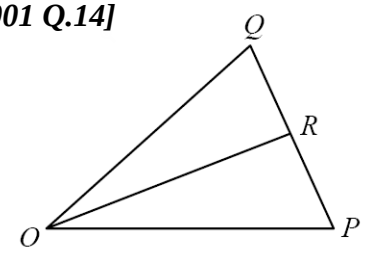


Figure 1(a)

(a) In Figure 1 (a). *OPQ* is a triangle. *R* is a point on *PQ* such that *PR : RQ = r:s.* Express $\overrightarrow{OR}$ in terms of *r. s,* $\overrightarrow{OP}$ and $\overrightarrow{OQ}$.

Hence show that if $\overrightarrow{OR} = m\overrightarrow{OP} + n\overrightarrow{OQ}$, then $m + n = 1$. ***(3 marks)***

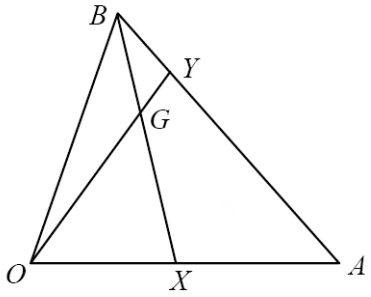


Figure 1(b)

(b) In Figure 1 (b). *QAB* is a triangle. *X* is the mid-point of *OA* and *Y* is a point on *B. BX* and *OY* intersect at point *G* where *BG* : *GX* = 1 : 3. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

(i) Express $\overrightarrow{OG}$ in terms of **a** and **b**.

(ii) Using (a), express $\overrightarrow{OY}$ in terms of **a** and **b**.
(Hint: Put $\overrightarrow{OY} = k\overrightarrow{OG}$.)

(iii) Moreover, *AG* is produced to a point *Z* on *OB*. Let $\overrightarrow{OZ} = h\overrightarrow{OB}$.

(1) Find the value of *h.*

(2) Explain whether *ZY* is parallel to *QA* or not.

(9 marks)

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2. ***[Add. Math 1999 Paper I Q.10]***

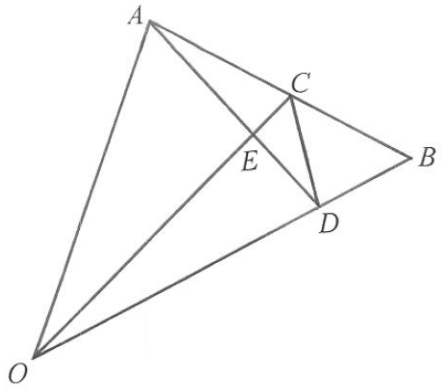


Figure 3

In Figure 3, *OAB* is a triangle. *C* and *D* are points on *AB* and *OB* respectively such that *AC : CB* = 8 : 7 and *OD* : *DB* = 16 : *5*. *OC* and *AD* intersect at a point *E*. Let $\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b**.

(a) Express $\overrightarrow{OC}$ and $\overrightarrow{AD}$ in terms of **a** and **b**. ***(4 marks)***

(b) Let $\overrightarrow{OE} = r\overrightarrow{OC}$ and $\overrightarrow{AE} = k\overrightarrow{AD}$.

(i) Express $\overrightarrow{OE}$ in terms of *r* ,**a** and **b**.

(ii) Express $\overrightarrow{OE}$ in terms of *k* ,**a** and **b**.

Hence show that $r = \frac{6}{7}$ and $k = \frac{3}{5}$. ***(6 marks)***

(c) It is given that *EC : ED* = 1 : 2. Using (b), or otherwise, find *EA* : *EO*. ***(3 marks)***

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3. ***[Add. Math 2005 Q.14]***

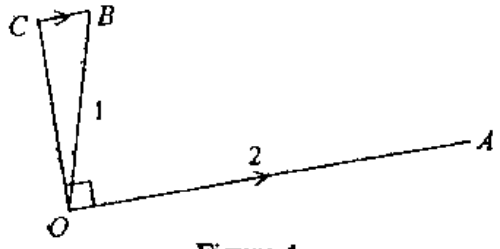


Figure 4

In Figure 4, $OA = 2$, $OB = 1$ and $\cos\angle AOB = \frac{1}{4}$. C is a point such that CB // OA and $OC \perp OA$. Let $\overrightarrow{OA}$ = **a**, $\overrightarrow{OB}$ = **b** and $\overrightarrow{OC}$ = **c**.

(a) Find $\overrightarrow{CB}$ in terms of **a**.

Hence, or otherwise, show that **c** = **b** – $\frac{1}{8}$**a**. ***(4 marks)***

(b)

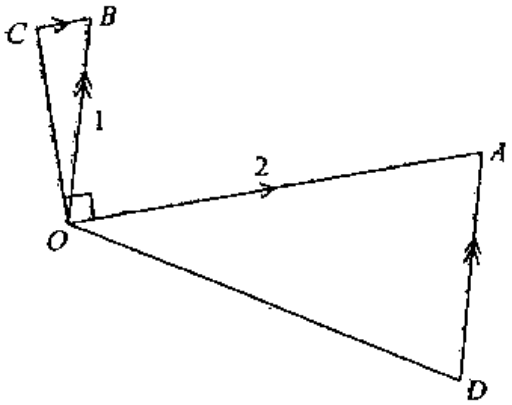


Figure 5

D is a point such that DA // OB and $OD = OA$ (see Figure 5). Let $\overrightarrow{OD}$ = **d**.

(i) By finding DA, or otherwise, express **d** in terms of **a** and **b**.

(ii) P is a point on the line segment CD such that $CP : PD = r : 1$. Express $\overrightarrow{OP}$ in terms of r, **a** and **b**

(iii) If M is the mid-point of AB, find the ratio in which OM divides CD. ***(8 marks)***

HKCEE Questions (level 3)

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